RESEARCH PROPOSAL

For Prospective MSc Thesis-Based Program

Interpretability and Robustness in Real-Time Computer Vision Systems

Motivation & Problem Statement

Deep learning models have achieved remarkable accuracy in image classification and pattern recognition. However, in real-world applications—autonomous systems, medical imaging, traffic control, financial analytics—accuracy alone is insufficient. Decision-makers need to understand why a model makes a specific prediction. This is the field of explainable AI (XAI). Yet despite progress, critical questions remain:

- How can we design vision systems that are simultaneously accurate, interpretable, and efficient for real-time deployment?
- What happens when models encounter distribution shifts—conditions different from training data? How do we anticipate and mitigate failures?
- How do we fuse information from multiple modalities (image + metadata) while preserving both accuracy and interpretability?

Personal Background & Motivation

I am a Data Scientist with 2+ years of professional experience in machine learning and computer vision systems. My journey toward these research questions began during my undergraduate Final Year Project (FYP), where I developed a hardware-based intelligent traffic management system using CNN-based vehicle detection.

While building this system, I encountered a humbling realization: achieving 92% classification accuracy did not guarantee a safe, deployable system. The model sometimes misclassified vehicles at dusk, or failed silently under heavy rain. When I tried to understand why, the model offered no insights—it was a black box. I began asking: How do we build vision systems that not only perform well, but fail gracefully and communicate their uncertainty?

Since then, my professional work has deepened this inquiry. I have designed and deployed multiple production AI systems:

- **Trading Assistant (Generative AI):** An LLM-powered system integrating multi-modal data sources (market feeds, sentiment analysis, SEC filings, insider activity) to generate financial insights. Challenge: How do we make LLM decisions transparent when the model fuses diverse, sometimes conflicting information?
- **News Clustering Pipeline:** An unsupervised learning system using BERT embeddings and HDBSCAN clustering to identify semantic patterns in high-volume

text data. Challenge: How robust are modern embeddings to noise? What patterns are actually meaningful?

- **Real-Time Data Systems:** Designed scalable pipelines (Kafka, PySpark) for processing complex datasets at production scale, introducing me to engineering trade-offs between accuracy, latency, and resource consumption.

Proposed Research Directions

I propose to explore three interconnected research questions during my Master's thesis:

1. Explainability Techniques for Real-Time Vision

Modern explainability methods (saliency maps, attention visualizations, LIME, SHAP) exist but are often computationally expensive or post-hoc. How can we embed explainability into the model design itself? Can we learn interpretable feature representations that preserve accuracy? What role can attention mechanisms play in designing inherently transparent models?

2. Robustness & Generalization Under Distribution Shift

Models trained on curated datasets often fail in the wild. How can we characterize and measure distribution shifts? Can we design models that either adapt online or provide reliable uncertainty estimates when they encounter novel data? How does explainability help us diagnose failure modes?

3. Multi-Modal Fusion with Interpretability

Real-world systems often integrate images with metadata, sensor data, or text. How do we design architectures that fuse these modalities while maintaining explainability? How do we know which modality the model relies on, and can we detect when it is overweighting one source?

Research Context & Alignment

The proposed research addresses critical gaps in current understanding of trustworthy AI systems. This work is particularly relevant for domains where both accuracy and interpretability are essential: medical imaging, autonomous systems, financial analytics, and critical infrastructure. My research builds on established work in explainable AI and extends it toward real-time, production-grade applications.

Key themes include:

- Explainable Deep Learning: Building transparent vision systems for high-stakes applications.
- Pattern Recognition & Generalization: Understanding and addressing robustness challenges when models encounter distribution shifts.
- Interpretability in Critical Applications: Developing methods where understanding failure modes and decision boundaries is paramount.

Expected Contributions

I aim to produce research that advances both theory and practice. Deliverables include:

- Peer-reviewed publications in computer vision and AI conferences or journals
- Open-source implementations of proposed methods
- Comprehensive experimental validation demonstrating improvements in interpretability, robustness, or generalization
- Thesis that meaningfully contributes to our understanding of trustworthy AI systems

Why This Matters

AI systems are increasingly deployed in high-stakes domains: medical diagnosis, autonomous driving, financial systems, criminal justice. In these contexts, accuracy alone is insufficient. We need models that are accurate, interpretable, robust, and honest about their limitations. This research bridges the gap between applied AI (which I have practiced professionally) and rigorous academic research that advances the field.

Conclusion

I am eager to transition my applied ML experience into fundamental research. I believe I can make meaningful contributions to explainable AI and robust vision systems. I am committed to producing publishable research while developing my skills as a researcher, and I look forward to collaborating with an advisor whose research aligns with these goals.